WiFi MIST Training

IEEE

802.15.1 Bluetooth BSIG

802.15.4 ZigBee (wireless light)
low power communication

IEEE 1394 old MAC firewire (iPod)
1394 AKA

802.3 → Ethernet
at → POE (amendments)

802.11 → Standard wireless

WiFi Alliance → Branding

Apple
Comcast
Sony
Zebel

Qualcomm
T-Mobile

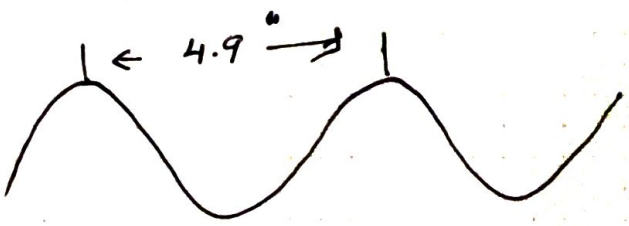
WiFi
WPA/WPA2/WPA3
WMM
WiFi-6

Ensures interoperability between vendors

2.4 & 5 GHz Bands

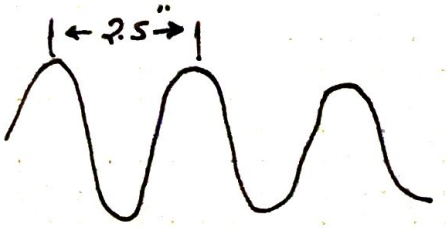
2.4

- low frequency
- longer wavelength
- Typically more range
- More crowded
- More non wifi interference
- 802.11 b/g/n/ax



5

High frequency
shorter wavelength
Less range
less crowded
Less non-WiFi interference
802.11 a/n/ac/ax



2.4 / 5 GHz → unlicensed spectrum

802.11 n → either 2.4 or 5 GHz

802.11 ac → only 5 GHz

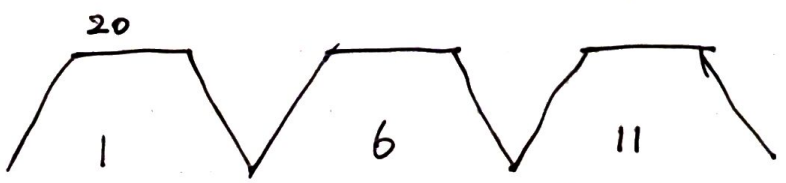
Major 802.11 Amendements

Amendments	Year	Frequency Band	Max Signal Rate	Modulation scheme	ch WD
802.11 prime (bar code)	1997	2.4	2 Mbps	DSSS	22MHz
802.11 b	1999	2.4	11Mbps	HR-DSSS	22MHz
802.11 a	1999	5	54	OFDM	20
802.11 g	2003	2.4	54	ERP OFDM	20
802.11 n Wi-Fi 4	2009	2.4 5	600 Mb	HT	20, 40
802.11 ac Wi-Fi 5	2013	5	6933 mbps	VHT	20, 40 80, 160
802.11 ax Wi-Fi 6		2.4 5	9600 mbps	HE	20, 40 80, 160

802.11 a - 90 feet but fast

802.11 b - 300 feet but slow

2.4 GHz - 11 channel 60 MHz



Non overlapping channels

co channel Interference

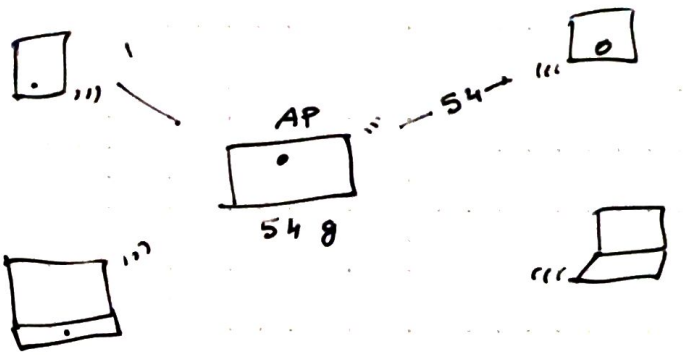
Half duplex

half duplex



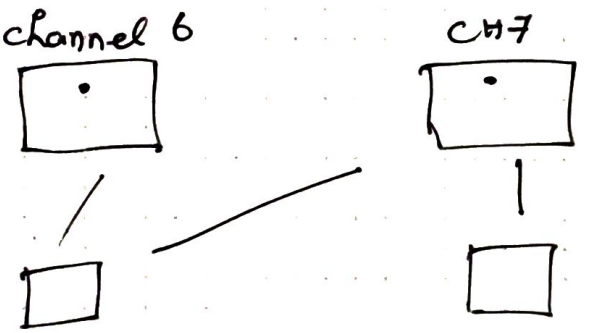
Slow devices on the channel consume more airtime.

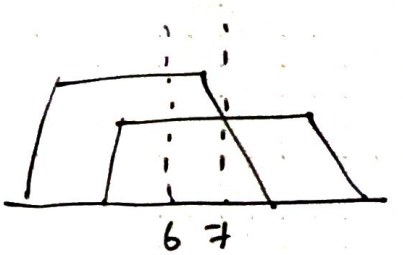
Data rate is based on signal strength



Adjacent channel Interference

STA's on overlapping channel will corrupt each other's transmission





MIST → 1, 6, 11

Channel reuse : A good channel plan minimizes co channel interferences.

HR- DSSS → 802.11b

CH 14 – Japan

5 GHz – 25 channels

UNII-1 36, 40, 44, 48

UNII-2 52, 56, 60, 64

UNII-2 Extended 100, 104, 108, 112, 116
120, 124, 128, 132, 136
140, 144

UNII-3 149, 153, 157
161

ISM 165

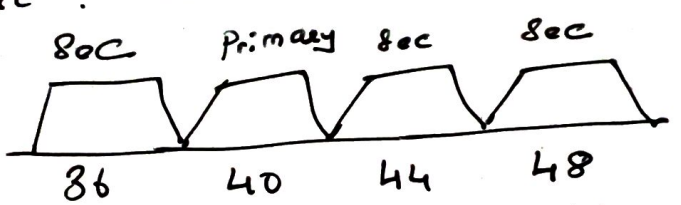
UNII-2/ Extended → DFS Dynamic frequency selection
Doppler weather radar exists in this channel.

Channel Bonding

introduced in 802.11n

802.11n : Two 20 MHz channels bonded to form 40 MHz

802.11ac : 20/40/80/160 MHz



2.4 GHz → No enough BW

5 GHz → 40 MHz in low density
20 MHz in high "
80 MHz never in enterprise
160 MHz not in practical

Signal strength

db → change in signal strength
dbm → absolute value

Rules of 3's & 10's

+3 dB → double the signal strength

−3 dB → Half the signal strength

+10 dB → 10 x more

−10 dB → 10 x less

dBm

−60 dBm 0.000001mW

−70 dBm 0.0000001 mw

−30 dBm – Good signal

−65 dBm → Voice app, mobile device

−67 dBm → Streaming video

−75 dBm → web browsing, email, instant messaging

−80 dBm → Generally reliable

−90 dBm → Drowning in the noise floor.

BSSID → mac addr of AP's radio

SSID → Friendly name of wifi n/w

ESSID → Multiple AP's in the wifi n/w

Other wireless technologies

BLE – 2.4 GHz, location services
Blue dot indoor navigation

Zigbee → 2.4 GHz, low power / throughput 250kbps. mesh n/w

CBRS → Citizen Broadband Radio service
3.5 – 3.7 GHz 5, 10, 15, 20, 40 MHz

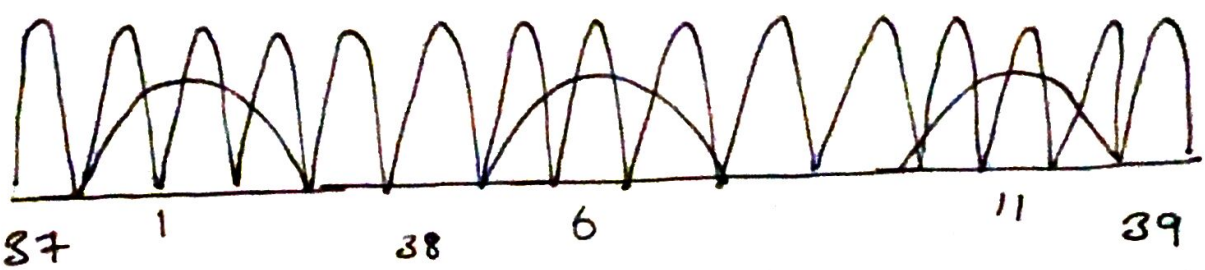
* Dedicated and tiered licensing model
* Upcoming/new technology
* Limited implementation in: Phone 11 / Google pixel Samsung S8

6 GHz

* Wifi in new spectrum
* Potential for LOT of new channel
* Upcoming/potential new technology

Bluetooth low energy

* Frequency hopping
* BLE uses bluetooth channel 37, 38, 39
* Channel sits between wifi channel 1, 6, 11
* BLE doesnot interfer with wifi



Ethernet

* CSMA/CD collision Detection
* only receives frames destined for it
* consistent data rate
* Throughput is line rate

Wireless NIC

* CSMA/CA collision Avoidance either listen or send/receive
* Demodulates all 802.11 frames it hears
* Data rate varies wildly based on signal strength SNR, traffic on the channel

* Actual throughput is roughly half the data rate
* Throughput decreases as the number of clients increases
* How to make Ethernet / wireless faster

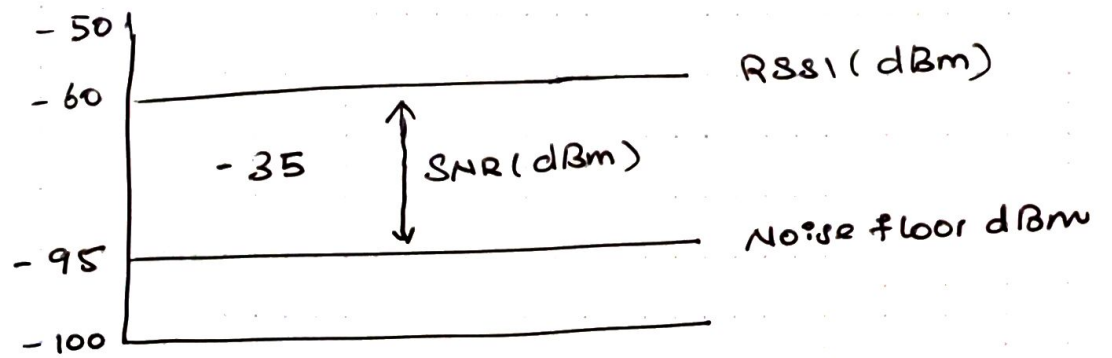
interference *

* decrease interference
* added shielding
* increase twist rate

Better signal strength

Signal to Noise ratio (SNR)

SNR → How much signal is there above the background noise.



Data rates:-

802.11b → 1, 2, 5.5, 11 mbps

802.11 a/g → 6, 9, 12, 18, 24, 36, 48, 54

802.11 n/ac → MCS index

http://mcsindex.com

https://www.wlanpros.com/mcs-index-charts

802.11 frame types:-

Management → STA associating / disassociating from wireless n/w

* Beacon
* Probe request
* Probe response
* Auth request
* Auth response
* Association request
* Disassociation response

Control :· controls RF medium and ensures that other frames are delivered.

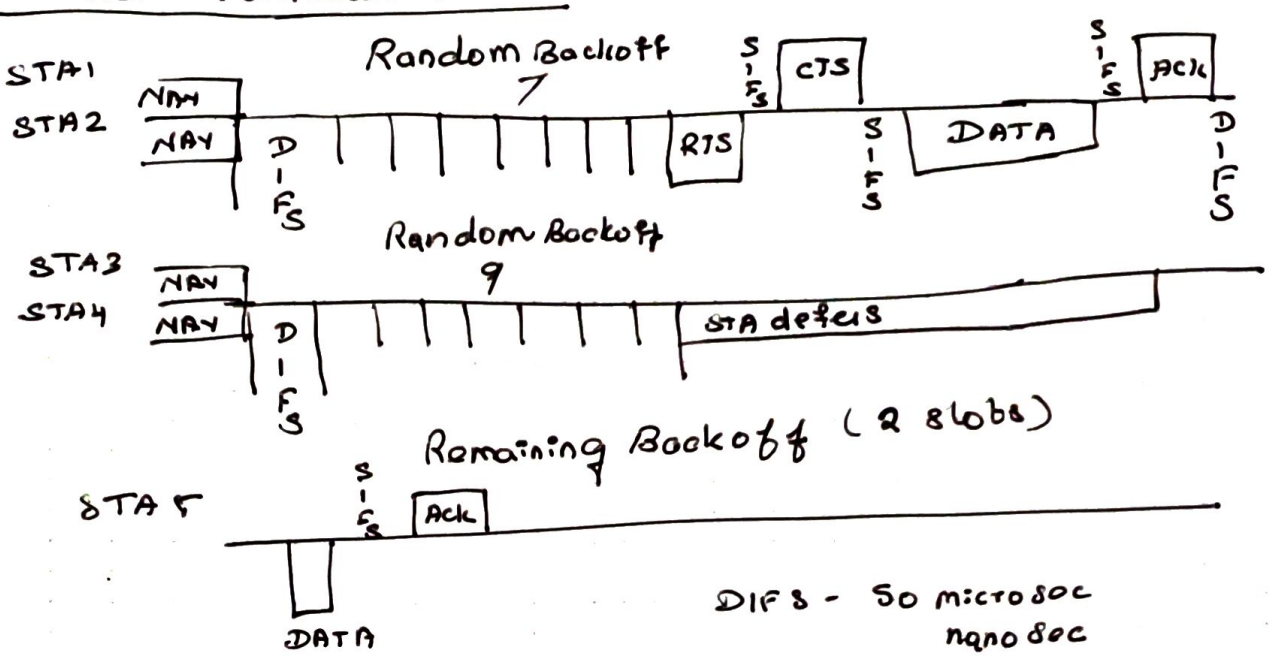
ACK, Block-ACK and RTS/CTS

DATA :· carries high level protocol data

DATA, CFDS, null data

Beacon → 10 times/sec

Airtime Arbitration Process



WiFi Roaming

Client Specific :·

Client decides when to roam

Client :· who to roam to

Green Diamond :· Concept by Keith Parson at Wireless LAN professionals.

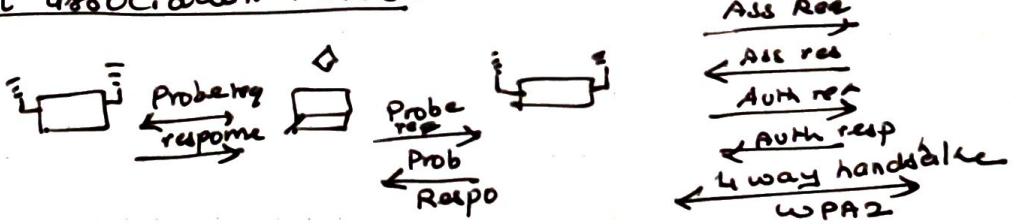
Decision matrix

Totally made up

SNR, signal strength

Retry rate

Client association Process



Wi-Fi Security

Open → No auth. and encryption

WEP (Wired equ Privacy) → Hex key (broken solution)

WPA → Wi-Fi Protected Access, intended as stop gap
54Mbps.

Two variants

WPA - personal / WPA-PSK
8 to 63 chara preshared passkey

WPA Enterprise
Credentials sent from AP to a Radius server

WPA 2 → New cipher suite
Required new hw

WPA 3 → Enhanced open
Encryption for open n/w with opportunistic wireless encryption (OWE)

WPA 3 personal - Provides foowand secrecy with simultaneous Auth of equals (SAE)

WPA3 Enterprise → stronger encryption

Wi-Fi Design

"If you aim at nothing, you'll hit it every time"
Z.g Z:gler

N/w Requirements:-

* Where do you need wi-fi
* where do you not need wi-fi
* what device you will be using
* how many device/user

* what type of application (eg) web, voice video etc

Client channel Support:-

* Not all clients support all channel
* AP's on unsupported channel won't be visible to the client
* https://clients.mikealbano.com

Determining Design requirements:-

1) Primary coverage:- min signal -67dBm
2) Secondary " : min -75 dBm
3) Minimum Data rate:- 24 mbps (not throughput)
4) SNR - how much background noise
   25 dB
5) channel overlap:- how many ap are allowed to share a channel above a certain signal
   Max 1 Ap above -85 dBm.

Ekahau Design

WiFi Basic-2

Review → half duplex (wireless)

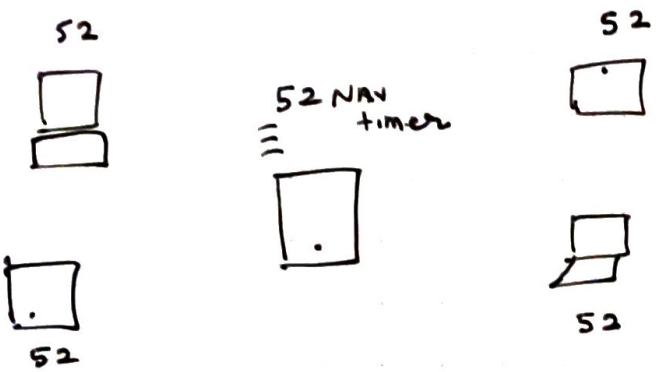
Physical carrier sense → to detect whether channel is busy or not

Energy detect:-
listens on channel to know anything talking on the channel
DO I hear RF on my channel
About -62 dBm threshold

Clear channel Assessment:-
DO I hear any other WiFi device on the channel
-82 dBm threshold (Backoff)

Virtual carrier sense::

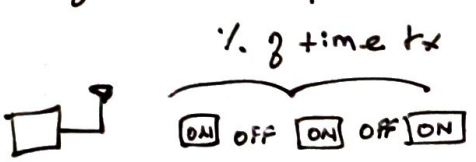
NAV → network Allocation Vector .3 sec (max)



tells other devices in the wifi n/w that amount of time device is going to use the channel

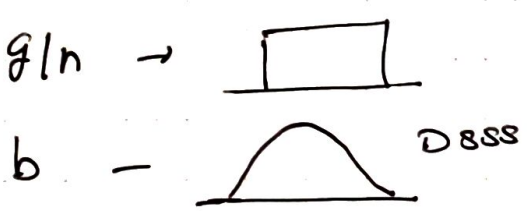
Duty cycle:: how often things are talking

* Layer1 management
* how often device spending talking / transmitting
* % of device spends talking



Utilization

* layer 1 measurement
* how often raw radio frequency activity found in the spectrum
* Spectrum analysis



Air time utilization → above layer 1

* Layer 2 measurement
* how much time 802.11 device reserved to talk
* measured with packet capture / analysis

nav - 1 - 32000 μs

Thing that consumes Airtime

* More devices (co channel interference)
* Poor coverage (device talks slower)
* legacy data rate
* Too many virtual SSID's

Multi Preshare key

* To avoid multiple SSID.
* We can define policy for each usage like printer etc with separate policy but using same SSID

Interesting facts about Site survey

* identify rouge antenna etc

SLE :. Service Level Expectation (MIST)

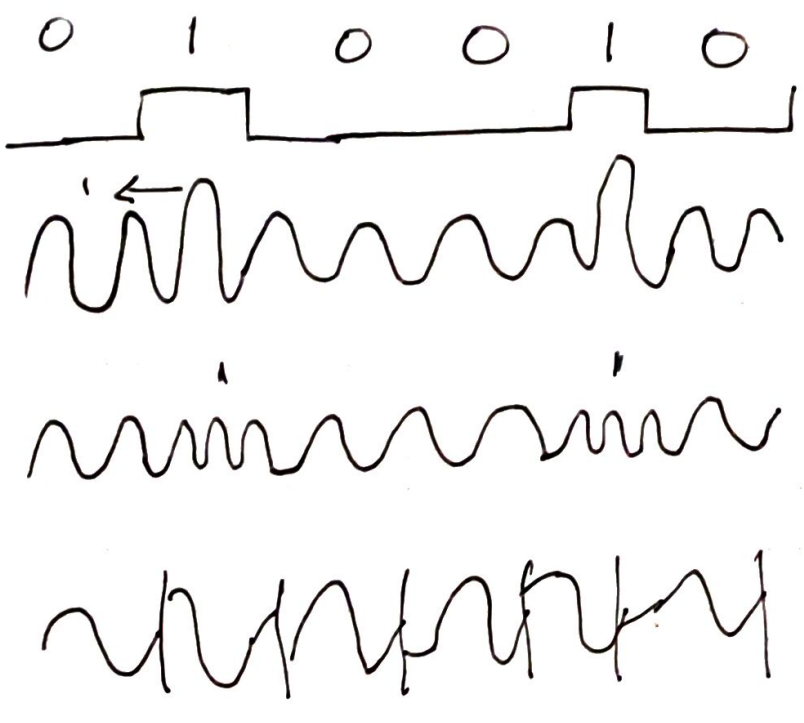
WiFi Basic - 3

DATA RATES

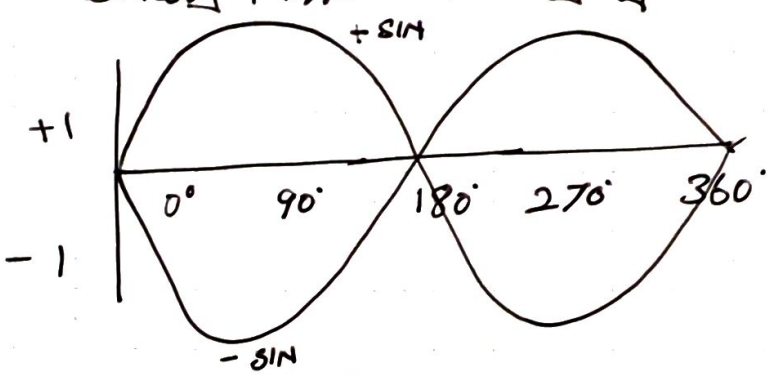
Data rate is influenced by below parameters

* Spatial stream
* Modulation
* Channel width
* SGI (Short Guard index) space between data transmit

Modulation :-



BPSK → Binary phase shift keying (simple modulation)

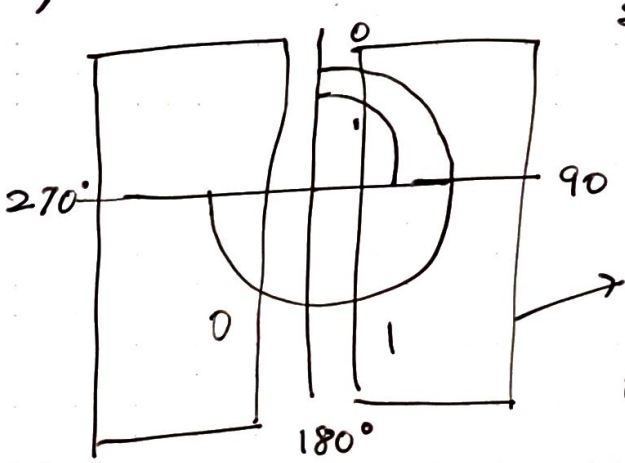


BSK

0 −SIN

1 SIN

Polar quardant chart



SIN where it peaks

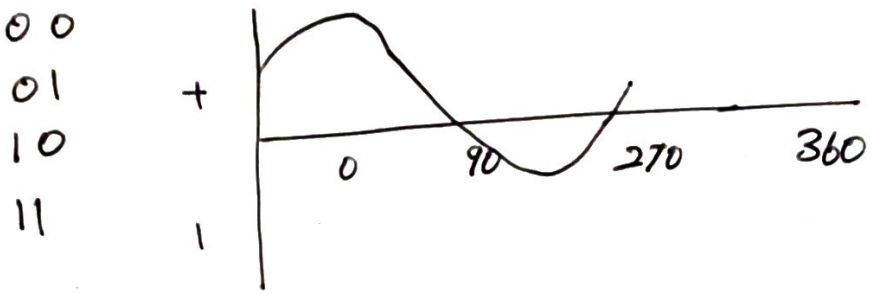
0 → peaks at 270°

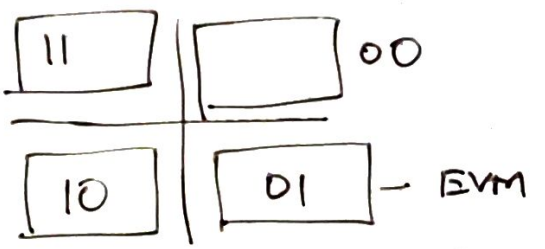
1 → " at 90°

→ EVM → Error vector magnitude

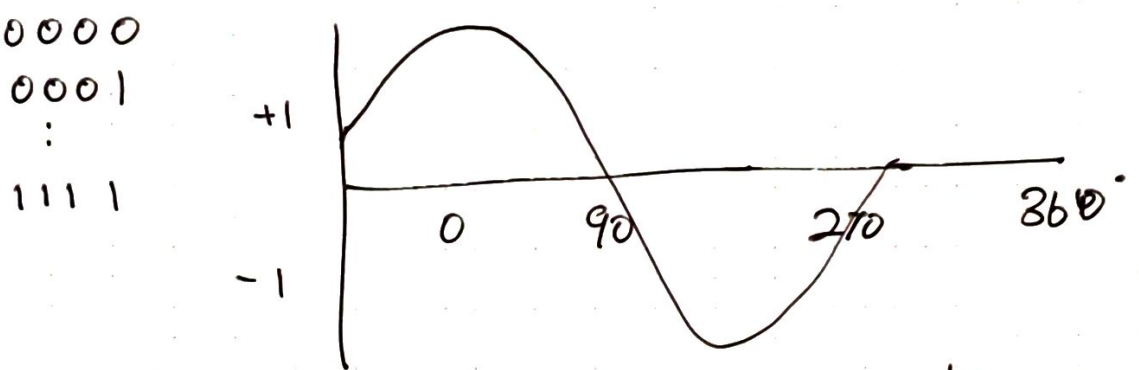
make the target land on 1 or 0

QPSK → Quaderature shift keying → double the data 2 bit at a time

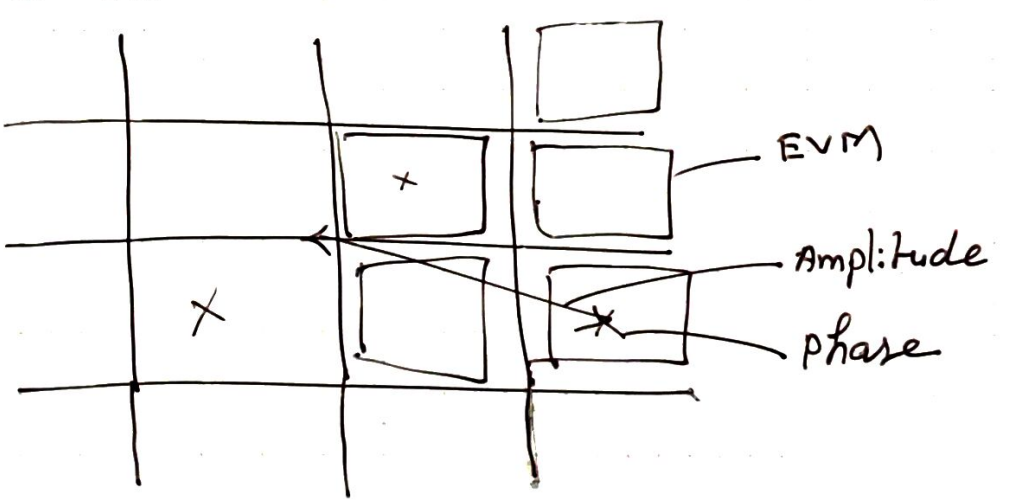




16 QAM → Phase and Amplitude modulation 4bit



* Phase represents where the wave peaks
* Amplitude represents how far the wave plotted from the center.



64 QAM - 6 bit to represent

000000
000001
⋮
111111

256 QAM 802.11ac 8 bits/symbol

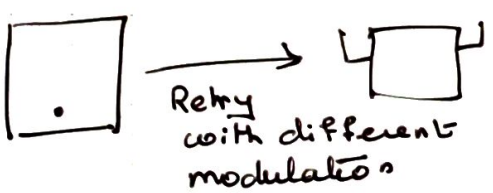
1024 QAM 802.11ax 10 bits/symbol

very close to AP, excellent data rate due to Qam

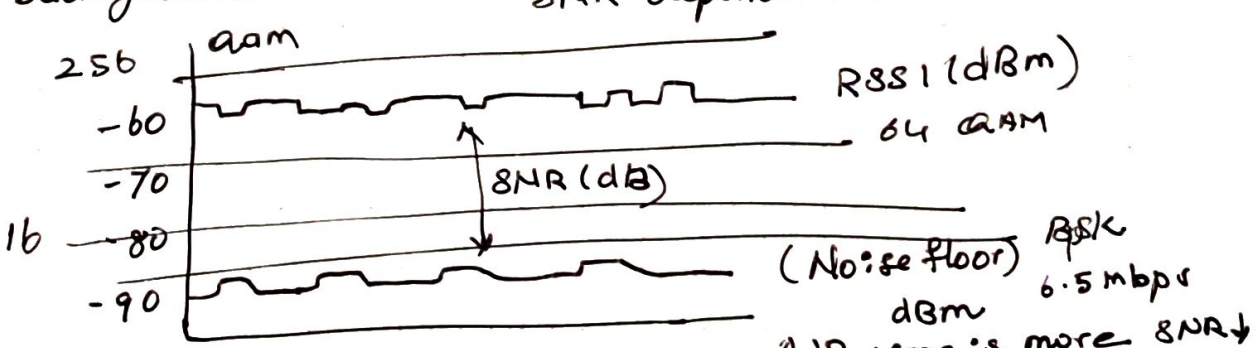
SNR ↓, rate shift Qam
signal strength
larger EVM box
less data rate

SNR is bad, then receiver stops sending ack

Data rate is based on retry



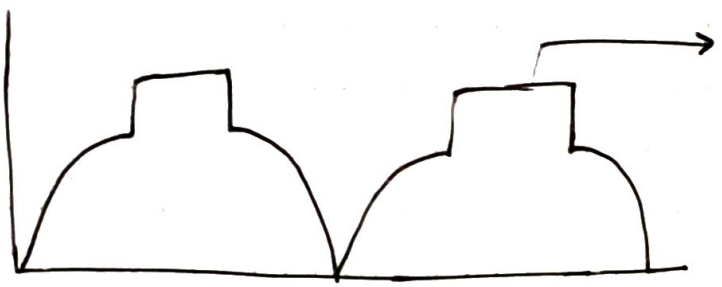
SNR → how much signal is there above the background noise. SNR depends on modulation

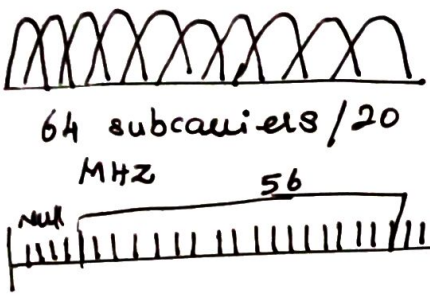


OFDM Sub carriers / Spatial Stream

802.11 a/g/n/ac/ax (to the extend) ⌐¬ table shape

802.11 b (DSSS) ⌒ curve shape





Every time, we transmit data we put symbol (x) on each one of the sub carrier

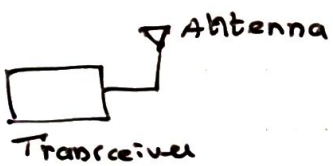
W.F: transceiver will decode different BPSK, QAM

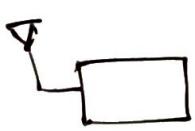
56 subcarriers
(4 pilot & 52 data)

64 312.5 KHZ
Null → No data
pilot → where to send
Data → string in the guiber

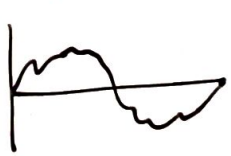
Spatial Streams

SISO → Single input Single output (1 radio / 1 Antenna)





Multipath → due to reflection, SISO device will be affect as it arrives in different / out of phase



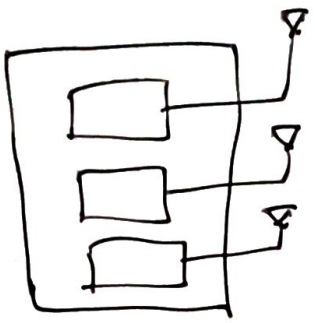
SISO with diversity (1 radio with 2 antenna)



any one antenna will be used

MIMO

Size : 8 the factor
battery life
Cost



3x3

2x2 - smaller
4x4 - high end AP

Advantage:- When transmits, receiving device on other end compares the signal, it take out of phase signal and it align them back which is called

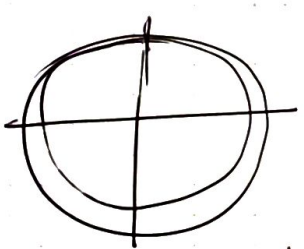
Maximum Ratio combining (MRC)

Result increase Amplitude

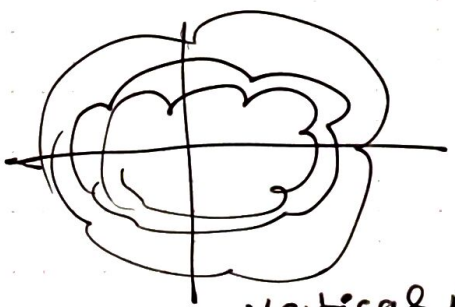
Spatial Multiplexing :- Faster data rate by putting different stream of data in different antenna/ reflection

MIMO and antenna Pattern

Antenna Pattern Do not use generic AP



Horizontal plane
Azimuth (Top down)



Vertical plane
Elevation
side view

Omnidirectional AP 43
↳ we can attach external antenna

Directional Patch Panel → Radiates RF in specific directional (warehouse, car parking)

High density Patch → 30° beam width

6 unique antenna inside external antenna

AP 43E Specification

badfi.com

2.4GHz : 4x4

5GHz : 4x4

Scanning : 2x2 (how many radio to perform RRM)

connectors : 6 dual-band

AP types and Architecture

1 Autonomous

* No central management
* No AP co ordination
* Traffic sent directly to the DS (Distribution system).

2. Lightweight AP

* Centrally managed
* AP coordinated
* Tunnels all traffic to the controller
* controller puts traffic on the DS

3. Cloud Managed

* cloud managed
* AP's coordinated by cloud
* Traffic sent directly to the DS
* Traffic sent to a Mist edge.

Mist Edge → functions as controller

Use case → controller based APs

Tunnel all traffic to mist edge like traditional controller.

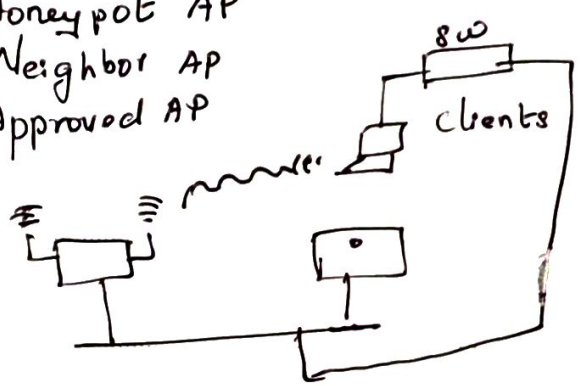
L2TP → end point but not controller

1701

Mist is microservices which scales automatically
↳ with AI which is different from other vendor

Mist config templates
config template
> High density data rate - min 18Mbps
> WLAN SSID
RF Template
2.4/5 GHz power

Network → Security
* Rogue AP — AP connected to same physical n/w
* Honeypot AP
* Neighbor AP
* Approved AP



clients connected to Rogue AP which is also connected to wired n/w

Honeypot → may not be connected in the n/w
advertise same SSID as MIST

RF template → Site configuration → config template

Mist RRM - Radio Resource Management

Network → Radio Management
AP density → how many APs can hear each other
(0.7 - 0.8) low coverage gap

Global RRM → AI driven
* Runs every date at 3 am local time / site "Scheduled RRM"
* Can be manually triggered / band "Triggered RRM"
* Global RRM is like the reset button

- Uses a large dataset accumulated over days to make decision
- Uses reinforcement learning.

Local RRM

- Runs on an ad-hoc basis
- Reacts to local device events
- cloud independent
- Event types
  - Auto channel selection
  - Auto triggered ACS
  - Interference AP co channel
  - Interference Non Wi-Fi
  - Interface co-channel external
    - Neighbor AP down / recover
    - Radar detected
    - Post-radar